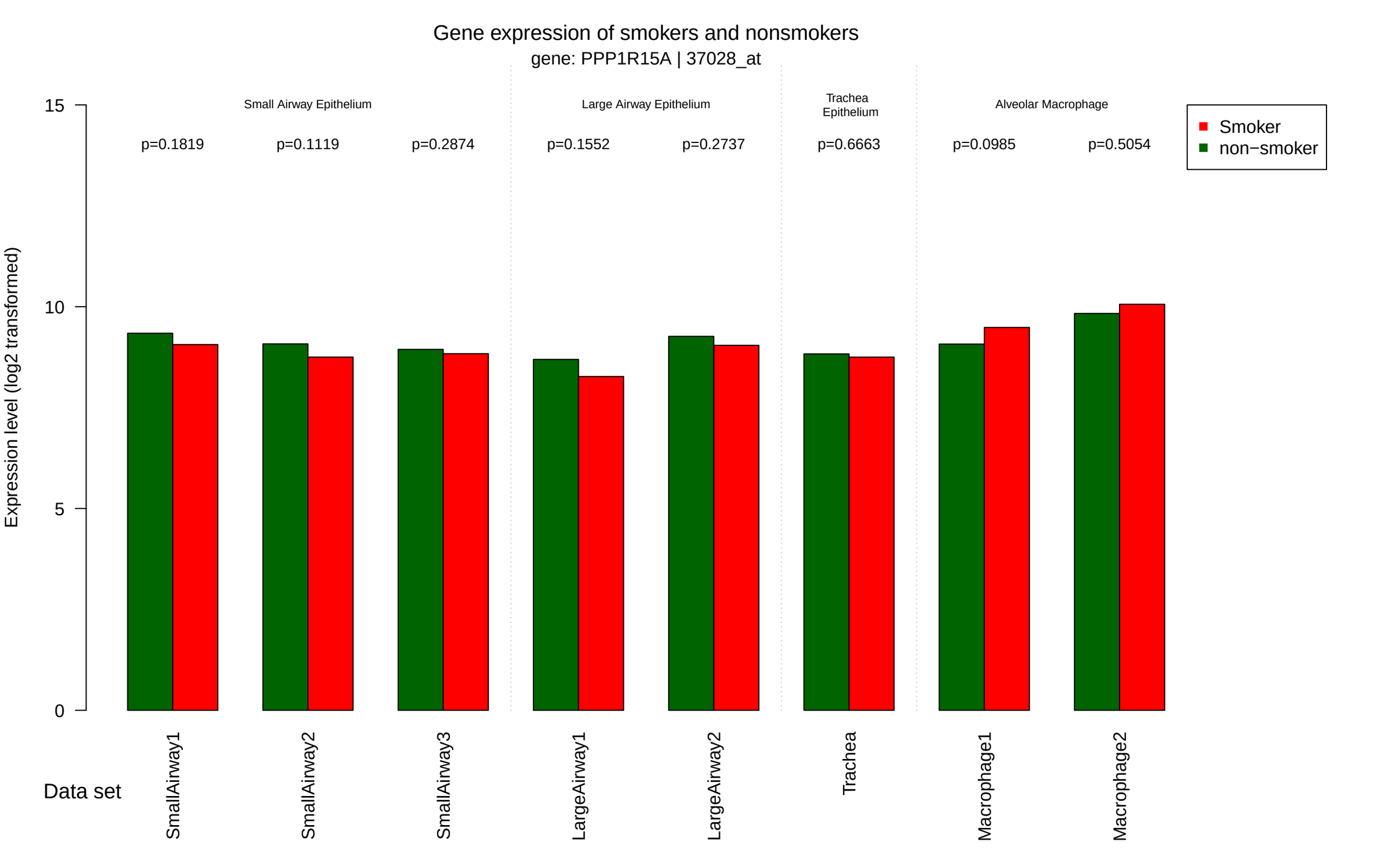
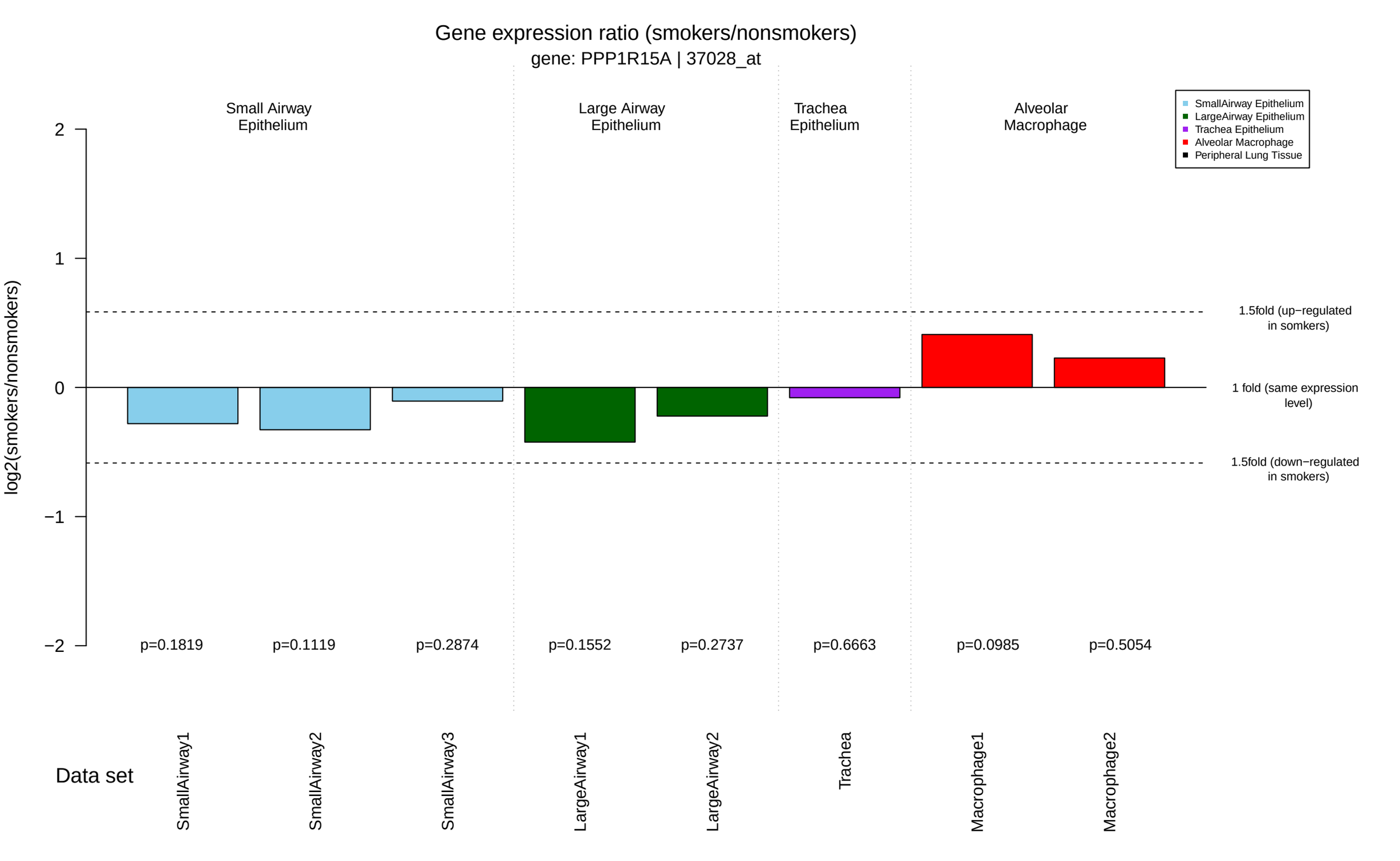
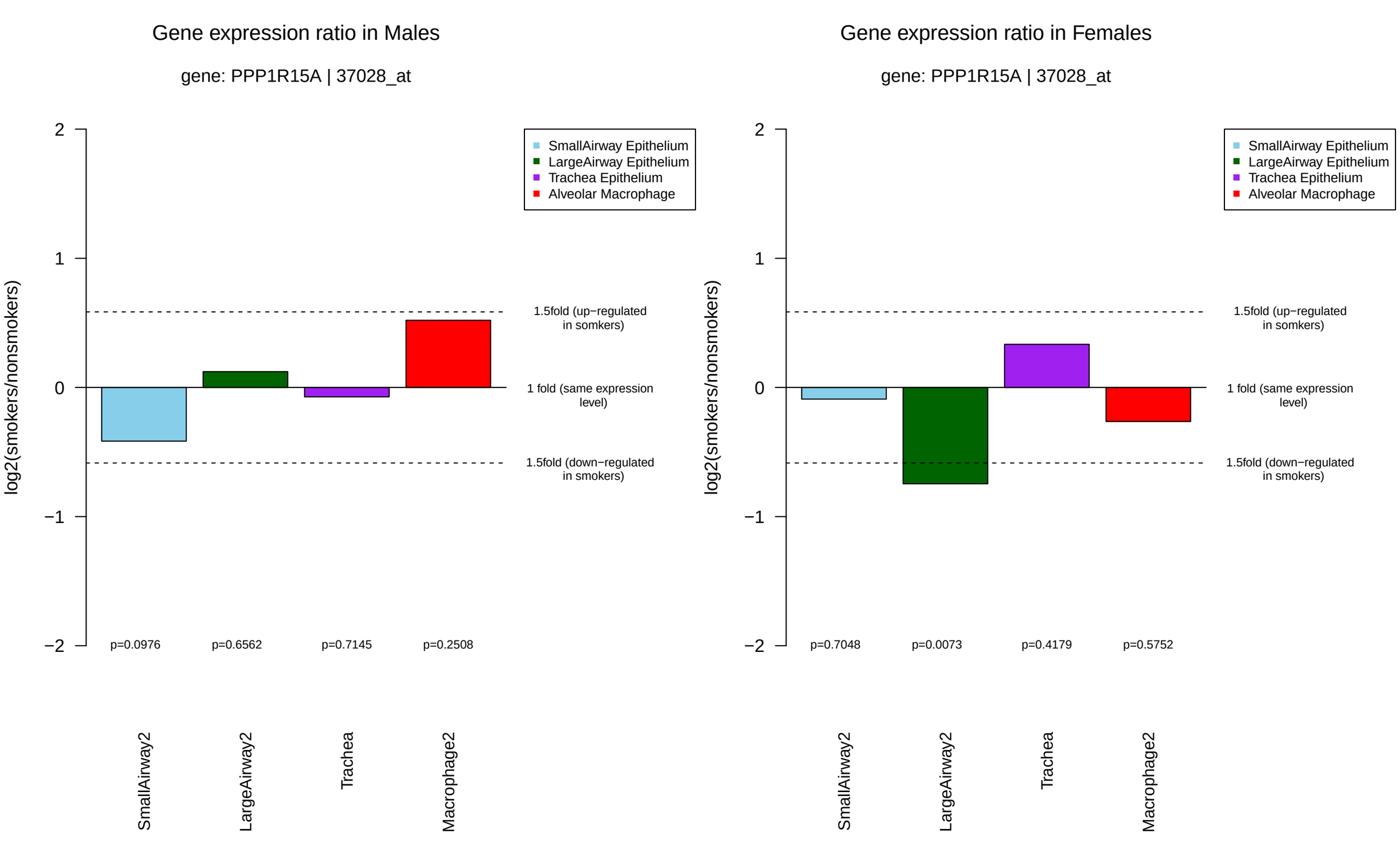
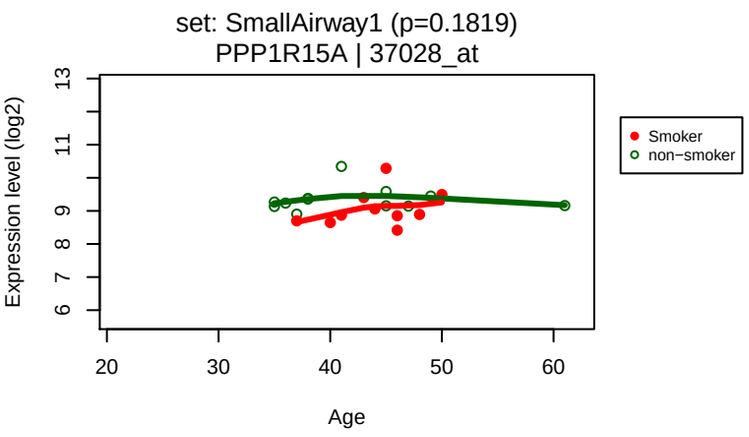
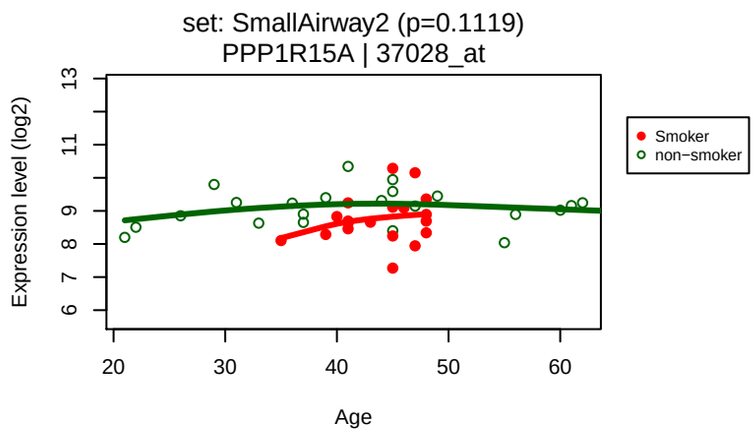
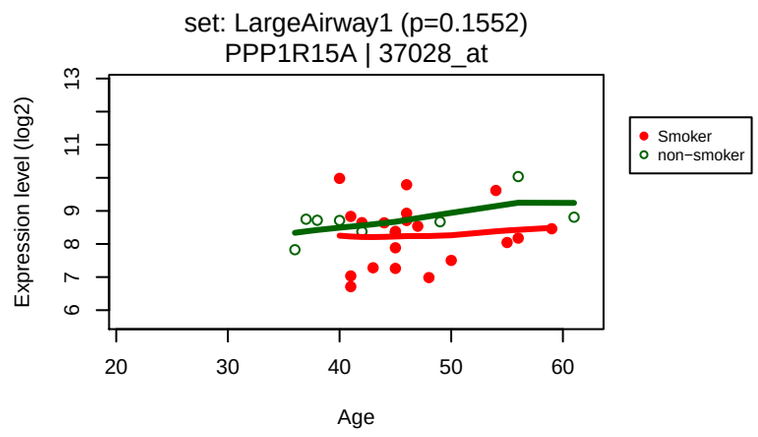

Gene expression of smokers and nonsmokers
gene: PPP1R15A | 37028_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1819
p=0.1119
p=0.2874
p=0.1552
p=0.2737
p=0.6663
p=0.0985
p=0.5054
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

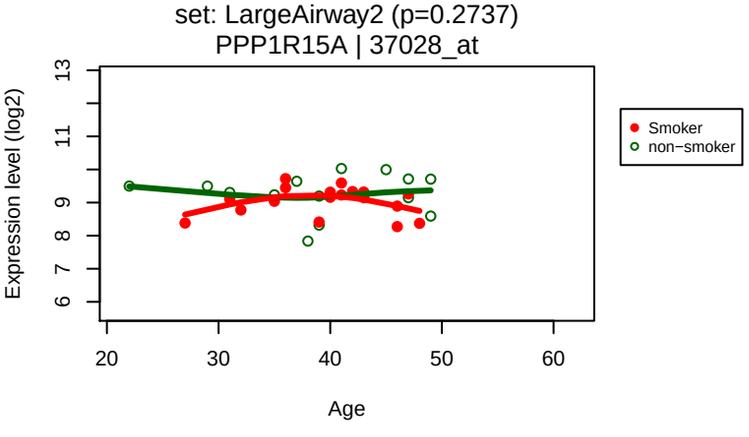

Gene expression ratio (smokers/nonsmokers)
gene: PPP1R15A | 37028_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1819
p=0.1119
p=0.2874
p=0.1552
p=0.2737
p=0.6663
p=0.0985
p=0.5054
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

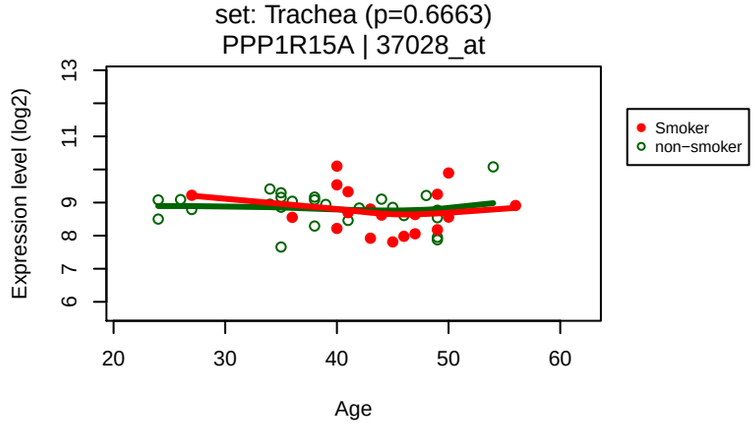

Gene expression ratio in Males
gene: PPP1R15A | 37028_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0976
p=0.6562
p=0.7145
p=0.2508
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PPP1R15A | 37028_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7048
p=0.0073
p=0.4179
p=0.5752
SmallAirway2
LargeAirway2
Trachea
Macrophage2

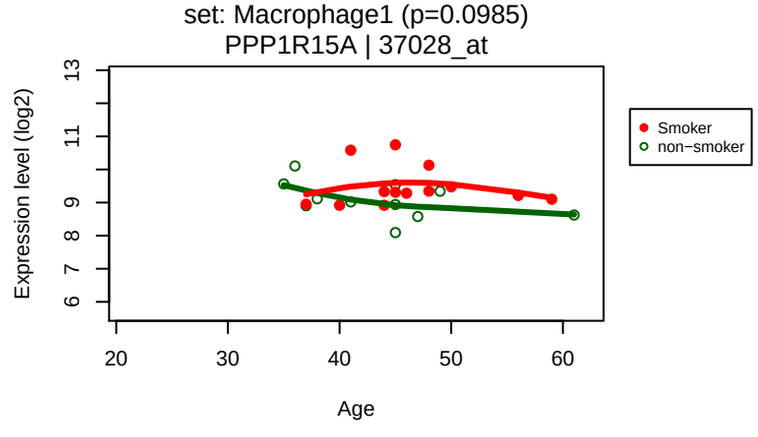

set: SmallAirway1 (p=0.1819)
PPP1R15A | 37028_at
Expression level (log2)
Age
Smoker
non−smoker

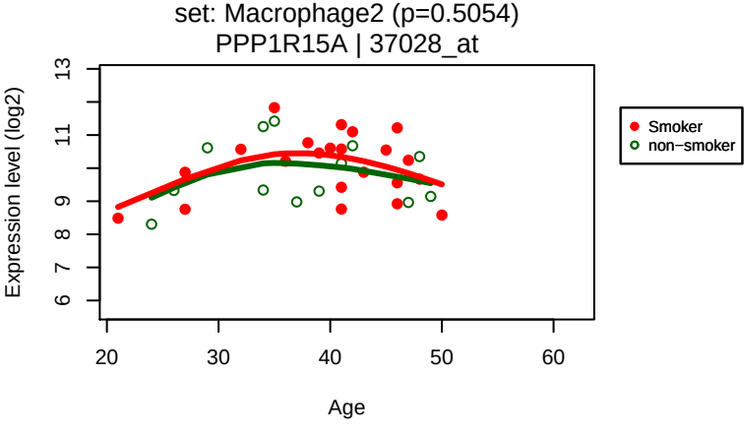

set: SmallAirway2 (p=0.1119)
PPP1R15A | 37028_at
Expression level (log2)
Age
Smoker
non−smoker

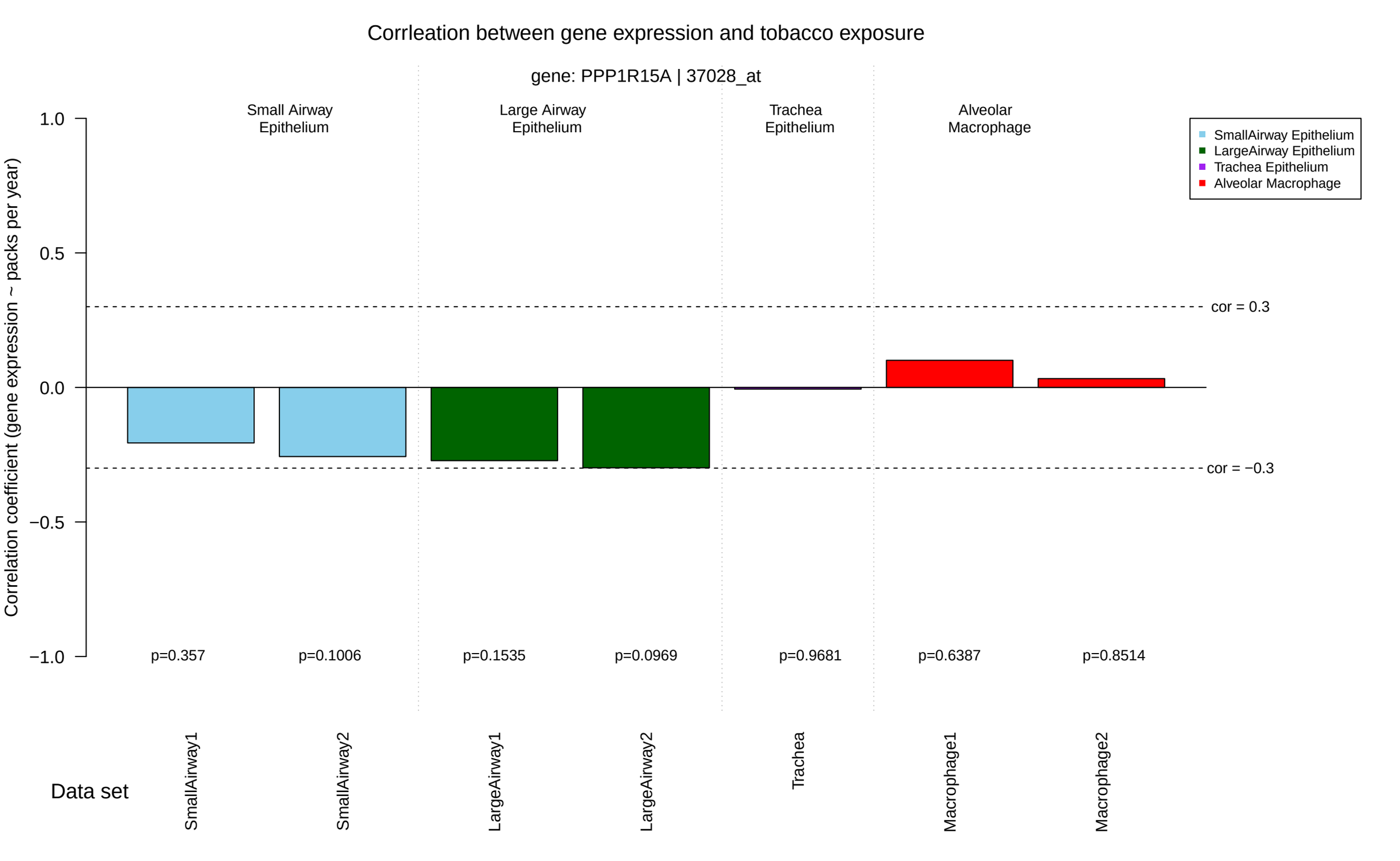

set: LargeAirway1 (p=0.1552)
PPP1R15A | 37028_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2737)
PPP1R15A | 37028_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.6663)
PPP1R15A | 37028_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0985)
PPP1R15A | 37028_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.5054)
PPP1R15A | 37028_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PPP1R15A | 37028_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.357
p=0.1006
p=0.1535
p=0.0969
p=0.9681
p=0.6387
p=0.8514
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2